Data Cleaning and Data Quality

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WEEK 2 LECTURE DECK

Data Cleaning and Data Quality

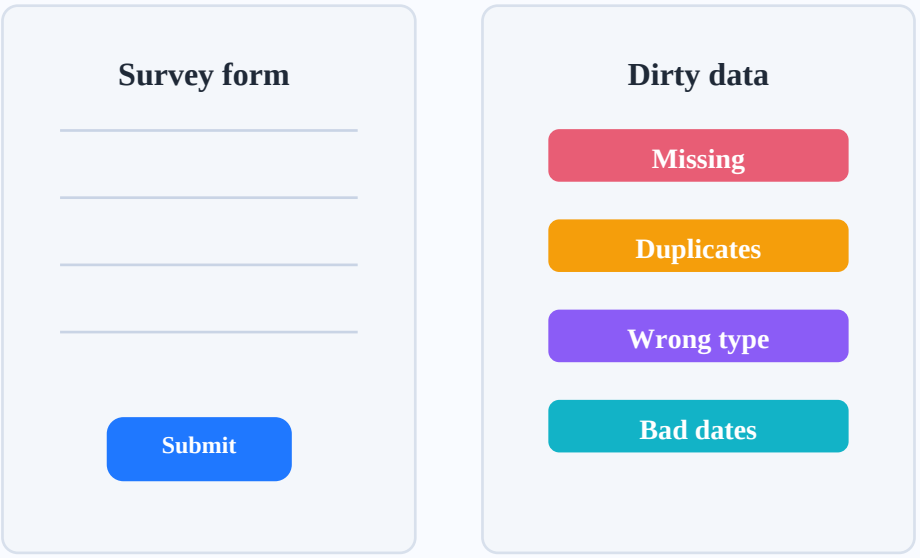
Topics for this week:

- data cleaning I: missing values, types, duplicates
- data quality checklist
- text normalization and date parsing
- export cleaned data to CSV

Sample project: Online Survey System

Clean survey data first, then trust the results.

Bad data creates bad charts and wrong decisions.



From raw responses to clean analytics

collect → check → clean → export → analyze

Week 2 roadmap

What students should understand and do by the end of this week

1. Missing values
2. Types
3. Duplicates
4. Quality checklist
5. Text cleaning
6. Date + export

Knowledge

Students can explain why missing values, wrong types, and duplicates create weak analysis.

Skills

Students can inspect a CSV file, clean key problems, normalize text, parse dates, and save a cleaned file.

Mindset

Do not trust raw data too quickly. Check first, clean second, analyze third.

Why clean data before analysis

Simple rule: clean first, then measure, then decide

Raw survey data

- Blank scores
- Mixed device names
- Repeated submissions
- Wrong date format

→

Cleaning work

- Fill or mark blanks
- Standardize text
- Remove duplicates
- Fix date values

→

Better output

- Correct KPIs
- Fair comparison
- Clear charts
- Better decisions

Example question for our survey project: “Which province has the lowest satisfaction score?”

If score values are missing or if one person submitted the form twice, the answer can become false.

Clean first

SESSION

Session 1: Data cleaning I and data quality

We start with the most common raw-data problems.

Missing values: what they mean and what to do

A blank cell is not always a small problem

Common reasons

- User skipped a question
- System did not save a value
- Wrong import from CSV
- Data type problem created a blank

Basic actions

- Keep as missing if it matters
- Fill with a safe value only when reasonable
- Remove the row if too much is missing
- Report how many missing values exist

Simple pandas check

```
df.isnull().sum()
df[df["score"].isnull()]
```

This shows how many missing values each column has, and which rows have a missing score.

Data types and duplicates

Wrong types confuse the computer; duplicate rows repeat the same story

Data type examples

score as text: "5" instead of number 5

submit_time as plain text instead of date

recommend as many forms: yes, Yes, Y, 1

Why this matters

- Text cannot always be averaged like numbers
- Dates cannot be sorted correctly if the format is broken
- Many forms of one answer create messy categories

Duplicate rows

In our survey project, duplicates may happen when:

- a user clicks submit twice
- the import script runs twice
- two files are combined badly

```
df.duplicated().sum()
df.drop_duplicates()
```

First count duplicate rows. Then remove them carefully, often by key fields such as response_id.

A small data quality checklist

Use the same checklist every time you open a new dataset

1. Columns
Are the column names clear and complete?

2. Missing values
Which columns have blanks, and how many?

3. Data types
Are numbers, text, and dates in the right form?

4. Duplicates
Are any rows repeated?

5. Categories
Do labels use one standard style?

6. Range check
Are values inside a normal range?

Tip: Run this checklist before you start KPI tables or charts.

SESSION

Session 2: Text cleaning, dates, and clean export

Now we fix values into a form that analysis can use.

Text normalization

Make similar values look the same

Before cleaning

mobile

Mobile

MOBILE

phone

smart phone

These may all mean almost the same thing, but the computer sees many categories.

→

After normalization

mobile

mobile

mobile

mobile

mobile

One standard form is easier to count, group, and compare.

Simple pandas idea

```
df["device"] = (
  df["device"].str.strip()
    .str.lower()
)
```

strip removes extra spaces. lower changes text to lowercase.

Date parsing

Dates should behave like dates, not only like text

Common raw date forms

2026-01-12

12/01/2026

Jan 12, 2026

2026/01/12 09:15

Mixed formats make sorting and filtering harder.

Why parse dates?

- Sort by time correctly
- Extract month, day, or hour later
- Filter recent records
- Find trends over time

Simple pandas idea

```
df["submit_time"] =
pd.to_datetime(df["submit_time"],
  errors="coerce")
```

If a date is broken, errors="coerce" turns it into a missing value so we can inspect it.

Export a cleaned CSV file

Keep raw data and cleaned data as two different files

Good workflow

- Load the raw survey CSV
- Inspect quality problems
- Clean missing values, text, and dates
- Save a new file, for example survey_responses_clean.csv

Why not overwrite the old file?

Keeping the raw file is safer. You can always go back and check what changed.

Simple export code

```
df.to_csv(
"survey_responses_clean.csv",
  index=False
)
```

Index = False avoids saving the extra row index column.

For our survey project

The cleaned CSV can be used to:

- build KPI tables
- create charts
- load into MySQL later
- support the Flask dashboard

Clean export is the bridge between data cleaning and real analytics work.

Week 2 lab activity

Use one small survey file and clean it step by step

In-class lab

- Open the sample survey CSV.
- Check missing values.
- Inspect data types.
- Count duplicate rows.
- Normalize the device column.
- Parse submit_time as a date.
- Save a clean CSV file.

Homework idea

- Write a short data quality report.
- Which columns had problems?
- What did you clean?
- What still needs discussion?

Teacher focus

- Watch for students who clean too fast without checking.
- Ask them to explain why they changed each value.

Week 2 recap, exit task, and next step

End this week with one clean file and one clear explanation

Key ideas

Missing values, wrong data types, duplicates, inconsistent text, and broken dates can damage analysis.

Exit task

Explain one cleaning action in your own words and show the line of code you used.

Next step

In Week 3, we will use cleaned data to compute summary statistics, group results, and build KPI tables.

Remember: *if the data is messy, the insight can also be messy. Clean data is the base of good analytics.*

```python
from flask import Flask, render_template
import pandas as pd

app = Flask(__name__)

def load_and_clean_data():
    # Load CSV
    df = pd.read_csv("survey_responses.csv")

    # Keep original row count
    original_rows = len(df)

    # Remove duplicate rows
    df = df.drop_duplicates()

    # Normalize text columns
    text_columns = ["province", "age_group", "device", "recommend"]
    for col in text_columns:
        if col in df.columns:
            df[col] = df[col].astype(str).str.strip().str.lower()

    # Standardize province names
    if "province" in df.columns:
        df["province"] = df["province"].replace({
            "phnom penh": "Phnom Penh",
            "siem reap": "Siem Reap",
            "kandal": "Kandal",
            "battambang": "Battambang",
            "kampot": "Kampot"
        })

    # Convert date column
    if "submit_time" in df.columns:
        df["submit_time"] = pd.to_datetime(df["submit_time"], errors="coerce")

    # Convert score to numeric
    if "score" in df.columns:
        df["score"] = pd.to_numeric(df["score"], errors="coerce")

    # Fill missing values
    if "score" in df.columns:
        df["score"] = df["score"].fillna(df["score"].median())

    if "device" in df.columns:
        df["device"] = df["device"].replace("nan", pd.NA)
        df["device"] = df["device"].fillna("unknown")

    if "recommend" in df.columns:
        df["recommend"] = df["recommend"].replace("nan", pd.NA)
        df["recommend"] = df["recommend"].fillna("unknown")

    if "province" in df.columns:
        df["province"] = df["province"].replace("nan", pd.NA)
        df["province"] = df["province"].fillna("Unknown")

    # Create data quality summary
    summary = {
        "original_rows": original_rows,
        "cleaned_rows": len(df),
        "duplicates_removed": original_rows - len(df),
        "missing_submit_time": int(df["submit_time"].isna().sum()) if "submit_time" in df.columns else 0,
        "missing_score": int(df["score"].isna().sum()) if "score" in df.columns else 0,
        "missing_device": int(df["device"].isna().sum()) if "device" in df.columns else 0,
    }

    return df, summary

@app.route("/")
def index():
    df, summary = load_and_clean_data()

    # Show first 10 rows
    table_html = df.head(10).to_html(classes="table table-bordered table-striped", index=False)

    return render_template("index.html", table=table_html, summary=summary)

if __name__ == "__main__":
    app.run(debug=True)
```

File name: app.py

templates > index.html

```html
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Survey Data Cleaning Dashboard</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 30px;
            background: #f4f6f9;
        }
        h1, h2 {
            color: #1f3c88;
        }
        .card {
            background: white;
            padding: 20px;
            margin-bottom: 20px;
            border-radius: 10px;
            box-shadow: 0 2px 8px rgba(0,0,0,0.08);
        }
        table {
            border-collapse: collapse;
            width: 100%;
            background: white;
        }
        table, th, td {
            border: 1px solid #ccc;
            padding: 8px;
        }
        th {
            background: #dbeafe;
        }
    </style>
</head>

<body>

<body>

    <h1>Online Survey System - Week 2 Flask Example</h1>

    <div class="card">
        <h2>Data Quality Summary</h2>
        <ul>
            <li><strong>Original rows:</strong> {{ summary.original_rows }}</li>
            <li><strong>Cleaned rows:</strong> {{ summary.cleaned_rows }}</li>
            <li><strong>Duplicates removed:</strong> {{ summary.duplicates_removed }}</li>
            <li><strong>Missing submit time:</strong> {{ summary.missing_submit_time }}</li>
            <li><strong>Missing score:</strong> {{ summary.missing_score }}</li>
            <li><strong>Missing device:</strong> {{ summary.missing_device }}</li>
        </ul>
    </div>

    <div class="card">
        <h2>Cleaned Survey Data</h2>
        {{ table|safe }}
    </div>

</body>
</html>
```

File name: templates/index.html

survey_responses.csv > data


```csv
response_id,submit_time,province,age_group,score,device,recommend,comment
1001,2026-01-12 09:10,Phnom Penh,18-24,5,mobile,yes,
1002,2026-01-12 09:15,Kandal,25-34,4,desktop,yes,
1003,2026-01-12 09:18,siem reap,18-24,2,mobile,no,
1004,2026-01-12 09:28,Battambang,35-44,,tablet,no,
1004,2026-01-12 09:28,Battambang,35-44,,tablet,no,
1005,2026/01/12 10:00, PHNOM PENH ,25-34,3,mobile,yes,
1006,,Kampot,18-24,4,,yes,
1007,2026-01-12 09:10,Phnom Penh,18-24,5,mobile,yes,Very good service
1008,2026-01-12 09:15,Kandal,25-34,4,desktop,yes,good
1009,2026-01-12 09:18,siem reap,18-24,2,mobile,no,too slow
1010,2026-01-12 09:28,Battambang,35-44,,tablet,no,not bad
1011,2026-01-12 09:28,Battambang,35-44,,tablet,no,not bad
1012,2026/01/12 10:00, PHNOM PENH ,25-34,3,mobile,yes,ok
1013,,Kampot,18-24,4,,yes,nice service
1014,2026-13-01 08:00,Banteay Meanchey,24-18,five,mobile,Yes,goooodd
1015,2026-01-12 11:00,Takeo,25-34,6,Desktop,maybe,excellent!!!!
1016,2026-01-12 11:15,SiemReap,18-24,1,mobile ,no,bad service??
1017,12-01-2026 11:20,Kandal,25_34,3,tablet,yes,it is fine
1018,2026-01-12 11:45,Phnom  Penh,18-24,-1,mobile,y,very late
1019,2026-01-12 12:00,Kep,N/A,4,MOBILE,n,good but expensive
1020,2026-01-12 12:10,kampot,35-44,3.5,desktop,yes,normal
1021,2026-01-12 12:30,Battambang,45-54,2,tablet,no,
```

File name: survey_responses.csv

Result:

127.0.0.1:5000

Online Survey System - Week 2 Flask Example

Data Quality Summary

- **Original rows:** 22
- **Cleaned rows:** 21
- **Duplicates removed:** 1
- **Missing submit time:** 6
- **Missing score:** 0
- **Missing device:** 0

Cleaned Survey Data

response_id	submit_time	province	age_group	score	device	recommend	comment
1001	2026-01-12 09:10:00	Phnom Penh	18-24	5.0	mobile	yes	NaN
1002	2026-01-12 09:15:00	Kandal	25-34	4.0	desktop	yes	NaN
1003	2026-01-12 09:18:00	Siem Reap	18-24	2.0	mobile	no	NaN
1004	2026-01-12 09:28:00	Battambang	35-44	3.5	tablet	no	NaN
1005	NaT	Phnom Penh	25-34	3.0	mobile	yes	NaN
1006	NaT	Kampot	18-24	4.0	unknown	yes	NaN
1007	2026-01-12 09:10:00	Phnom Penh	18-24	5.0	mobile	yes	Very good service
1008	2026-01-12 09:15:00	Kandal	25-34	4.0	desktop	yes	good
1009	2026-01-12 09:18:00	Siem Reap	18-24	2.0	mobile	no	too slow
1010	2026-01-12 09:28:00	Battambang	35-44	3.5	tablet	no	not bad

Thanks for attention,

Any Question?